

Q1 - 24 January - Shift 1

Assume that protons and neutrons have equal masses. Mass of a nucleon is 1.6×10^{-27} kg and radius of nucleus is $1.5 \times 10^{-15} A^{1/3}$ m. The approximate ratio of the nuclear density and water density is $n \times 10^{13}$. The value of n is _____.

Space for your notes:

Q2 - 24 January - Shift 2

A photon is emitted in transition from $n = 4$ to $n = 1$ level in hydrogen atom. The corresponding wavelength for this transition is (given, $h = 4 \times 10^{-15}$ eVs) :

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- (1) 94.1 nm
- (2) 941 nm
- (3) 97.4 nm
- (4) 99.3 nm

Q3 - 25 January - Shift 1

The wavelength of the radiation emitted is λ_0 when an electron jumps from the second excited state to the first excited state of hydrogen atom. If the electron jumps from the third excited state to the second orbit of the hydrogen atom, the wavelength of the radiation emitted will be $\frac{20}{x} \lambda_0$. The value of x is _____.

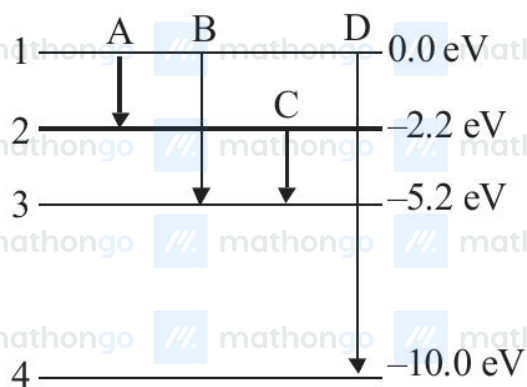
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Q4 - 25 January - Shift 2

The energy levels of an atom is shown is figure.

Which one of these transitions will result in the emission of a photon of wavelength 124.1 nm ?

Given ($h = 6.62 \times 10^{-34}$ Js)



(1) B

(2) A

(3) C

(4) D

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Q5 - 30 January - Shift 1

Speed of an electron in Bohr's 7th orbit for Hydrogen atom is 3.6×10^6 m/s. The corresponding speed of the electron in 3rd orbit, in m/s is :

(1) (1.8×10^6)

(2) (7.5×10^6)

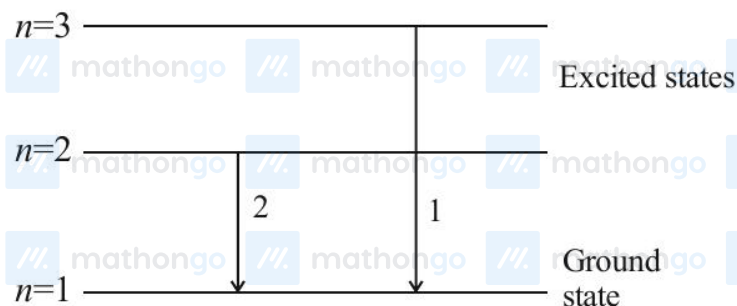
(3) (3.6×10^6)

(4) (8.4×10^6)

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Q6 - 31 January - Shift 1

For hydrogen atom, λ_1 and λ_2 are the wavelengths corresponding to the transitions 1 and 2 respectively as shown in figure. The ratio of λ_1 and λ_2 is $\frac{x}{32}$. The value of x is _____.



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Q7 - 31 January - Shift 2

The radius of electron's second stationary orbit in Bohr's atom is R . The radius of 3rd orbit will be

- (1) $\frac{R}{3}$ (2) $2.25R$
(3) $3R$ (4) $9R$

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Q8 - 31 January - Shift 2

If the binding energy of ground state electron in a hydrogen atom is 13.6 eV , then, the energy required to remove the electron from the second excited state of Li^{2+} will be: $x \times 10^{-1} \text{ eV}$. The value of x is _____.

Space for your notes:

Q9 - 01 February - Shift 1

The mass of proton, neutron and helium nucleus are respectively 1.0073 u, 1.0087 u and 4.0015u.

The binding energy of helium nucleus is:

- (1) 14.2 MeV (2) 28.4 MeV
(3) 56.8 MeV (4) 7.1 MeV

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Q10 - 01 February - Shift 1

A light of energy 12.75 eV is incident on a hydrogen atom in its ground state. The atom absorbs the radiation and reaches to one of its excited states. The angular momentum of the atom

in the excited state is $\frac{x}{\pi} \times 10^{-17}$ eVs. The value of x is _____ (use $h = 4.14 \times 10^{-15}$ eVs, $c = 3 \times 10^8$ ms⁻¹).

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Q11 - 01 February - Shift 2

An electron of a hydrogen like atom, having $Z = 4$, jumps from 4th energy state to 2nd energy state, The energy released in this process, will be:

(Given $R_{ch} = 13.6$ eV)

Where R = Rydberg constant

c = Speed of light in vacuum

h = Planck's constant

(1) 13.6 eV

(2) 10.5 eV

(3) 3.4 eV

(4) 40.8 eV

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Q12 - 01 February - Shift 2

Nucleus a having $Z = 17$ and equal number of protons and neutrons has 1.2 MeV binding energy per nucleon.

Another nucleus B of $Z = 12$ has total 26 nucleons and 1.8 MeV binding energy per nucleons.

The difference of binding energy of B and A will be _____ MeV.

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Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (11)

Q2 (1)

Q3 (27)

Q4 (4)

Q5 (4)

Q6 (27)

Q7 (2)

Q8 (136)

Q9 (2)

Q10 (828)

Q11 (4)

Q12 (6)

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Q1 (11)

$$\text{density of nuclei} = \frac{\text{mass of nuclei}}{\text{volume of nuclei}}$$

$$\rho = \frac{1.6 \times 10^{-27} \text{ A}}{\frac{4}{3} \pi (1.5 \times 10^{-15})^3 \text{ A}}$$

$$= \frac{1.6 \times 10^{-27}}{14.14 \times 10^{-45}} = 0.113 \times 10^{18}$$

$$\rho_w = 10^3$$

$$\frac{\rho}{\rho_w} = 11.31 \times 10^{13}$$

Hence ρ_w

Q2 (1)

$$\frac{hc}{\lambda} = \left[1 - \frac{1}{16} \right] (13.6 \text{ eV})$$

$$\text{So, } \lambda = 94.1 \text{ nm}$$

Q3 (27)

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$$\begin{array}{r} n=4 \\ \hline n=3 \\ \hline n=2 \\ \hline n=1 \end{array}$$

Second excited state $\rightarrow$ first excited state

$$n=3 \rightarrow n=2$$

$$\frac{hc}{\lambda_0} = 13.6 \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \dots\dots (i)$$

Third excited state $\rightarrow$ second orbit

$$n=4 \rightarrow n=2$$

$$\frac{hc}{(20\lambda_0 / x)} = 13.6 \left(\frac{1}{2^2} - \frac{1}{4^2} \right) \dots\dots (ii)$$

$$(ii) \div (i)$$

$$\frac{x}{20} = \frac{\frac{1}{2^2} - \frac{1}{4^2}}{\frac{1}{2^2} - \frac{1}{3^2}}$$

$$x = 27$$

Q4 (4)

$$\lambda = \frac{hc}{\Delta E}$$

$$\Delta E_A = 2.2 \text{ eV}$$

$$\Delta E_B = 5.2 \text{ eV}$$

$$\Delta E_C = 3 \text{ eV}$$

$$\Delta E_D = 10 \text{ eV}$$

$$\lambda_A = \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{2.2 \times 1.6 \times 10^{-19}}$$

$$= \frac{12.41 \times 10^{-7}}{2.2} \text{ m}$$

$$= \frac{1241}{2.2} \text{ nm} = 564 \text{ nm}$$

$$\lambda_B = \frac{1241}{5.2} \text{ nm} = 238.65 \text{ nm}$$

$$\lambda_C = \frac{1241}{3} \text{ nm} = 413.66 \text{ nm}$$

$$\lambda_D = \frac{1241}{10} = 124.1 \text{ nm}$$

Q5 (4)

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$$V_n \propto \frac{Z}{n}$$

$$Z = 1, \therefore V_n \propto \frac{1}{n}$$

$$\therefore \frac{V_3}{V_7} = \frac{7}{3}$$

$$\therefore V_3 = \frac{7}{3} V_7$$

$$= \frac{7}{3} \times 3.6 \times 10^6 \text{ m/s}$$

$$= 8.4 \times 10^6 \text{ m/s}$$

Q6 (27)

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$$\frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\frac{1}{\lambda_1} = RZ^2 \left[\frac{1}{1^2} - \frac{1}{3^2} \right] = \frac{8}{9} RZ^2 \dots\dots\dots (1)$$

$$\frac{1}{\lambda_2} = RZ^2 \left[\frac{1}{1^2} - \frac{1}{2^2} \right] = \frac{3}{4} RZ^2 \dots\dots\dots (2)$$

$$\frac{1}{2} \Rightarrow \frac{\lambda_2}{\lambda_1} = \frac{8}{9} \times \frac{4}{3} = \frac{32}{27}$$

$$\frac{\lambda_1}{\lambda_2} = \frac{27}{32}$$

Ans. 27

Q7 (2)

No Solution

Q8 (136)

No Solution

Q9 (2)

$$B.E \text{ of Helium} = (2m_p + 2m_n - m_{He}) c^2$$

$$= 28.4 \text{ MeV}$$

Q10 (828)

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In the ground state energy = -13.6 eV

So energy

$$\frac{-13.6 \text{ eV}}{n^2} = -13.6 + 12.75$$

$$\frac{-13.6 \text{ eV}}{n^2} = -0.85$$

$$n = \sqrt{16}$$

$$\boxed{n=4}$$

$$\text{Angular momentum} = \frac{nh}{2\pi} = \frac{4h}{2\pi} = \frac{2h}{\pi}$$

$$\text{Angular momentum} = \frac{2}{\pi} \times 4.14 \times 10^{-15}$$

$$= \frac{828 \times 10^{-17}}{\pi} \text{ eVs}$$

Q11 (4)

$$\Delta E = 13.6 Z^2 \left[\frac{1}{2^2} - \frac{1}{4^2} \right] \text{ eV}$$

$$= 13.6 \times (4)^2 \left(\frac{1}{4} - \frac{1}{16} \right) \text{ eV.}$$

$$= 13.6 [4 - 1] \text{ eV}$$

$$= 13.6 \times 3 = 40.8 \text{ eV}$$

Q12 (6)

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Total binding energy = $1.2 \times 34 = 40.8$ MeV

For B mass number = 26

total binding energy = 1.8×26 MeV

= 46.8 MeV

Difference of BE = 6 MeV

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